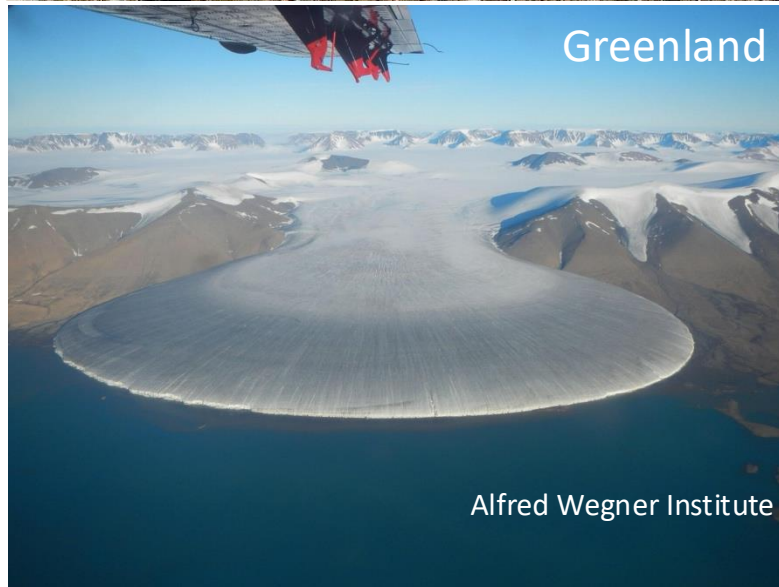
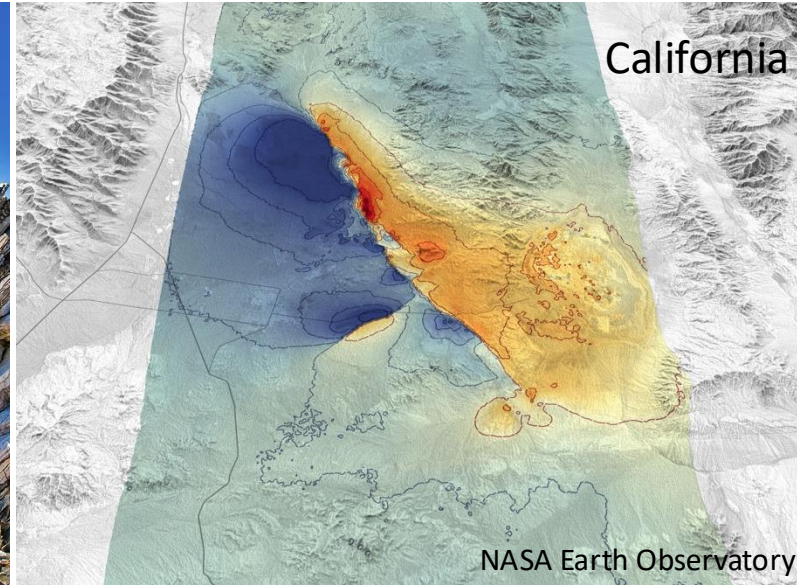


GEO 325C/398C Continuum Mechanics

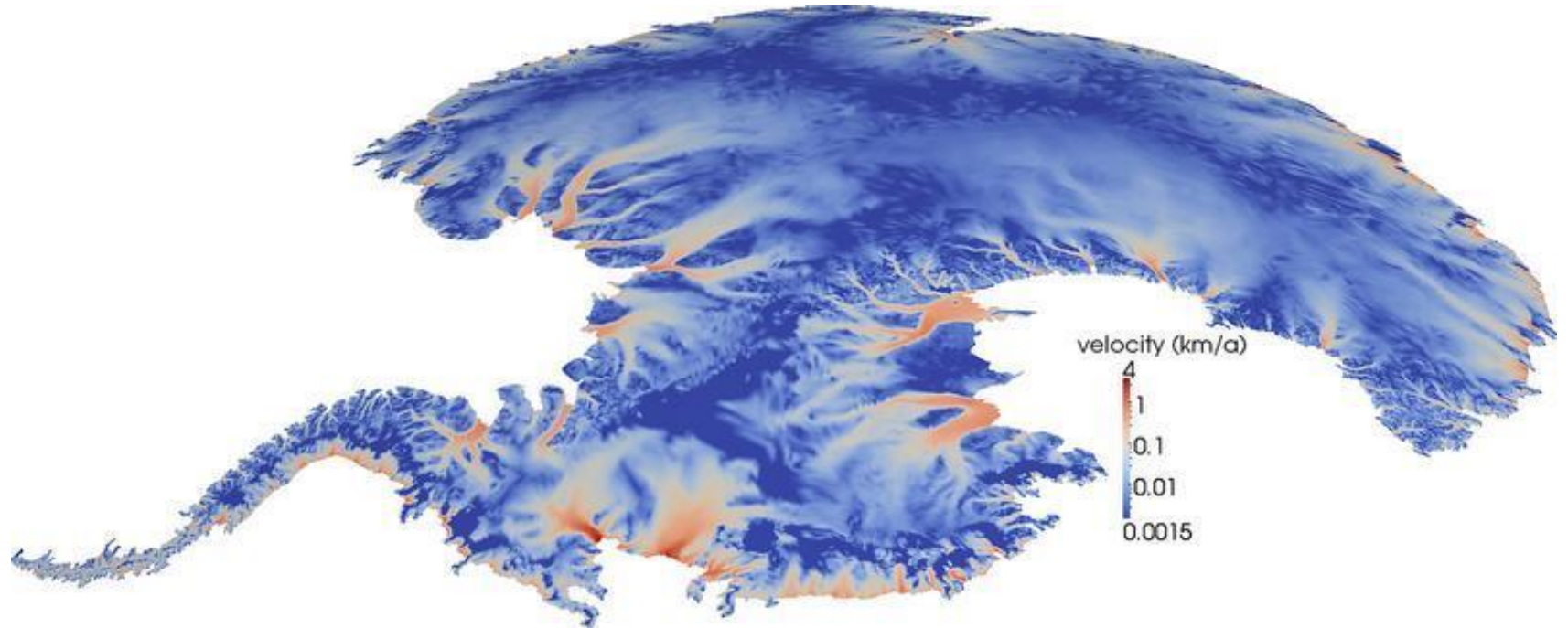


Why take this class?

This class provides the foundation for the modeling of all* **dynamical processes** in the Earth Sciences. It is excellent preparation for higher division classes in geodynamics, seismology, geomechanics, glaciology, atmospheric and ocean modeling.

Numerical model of the Antarctic ice sheet by former UT graduate student Tobin Isaac, taken from the [website of the ODEN Institute](#).

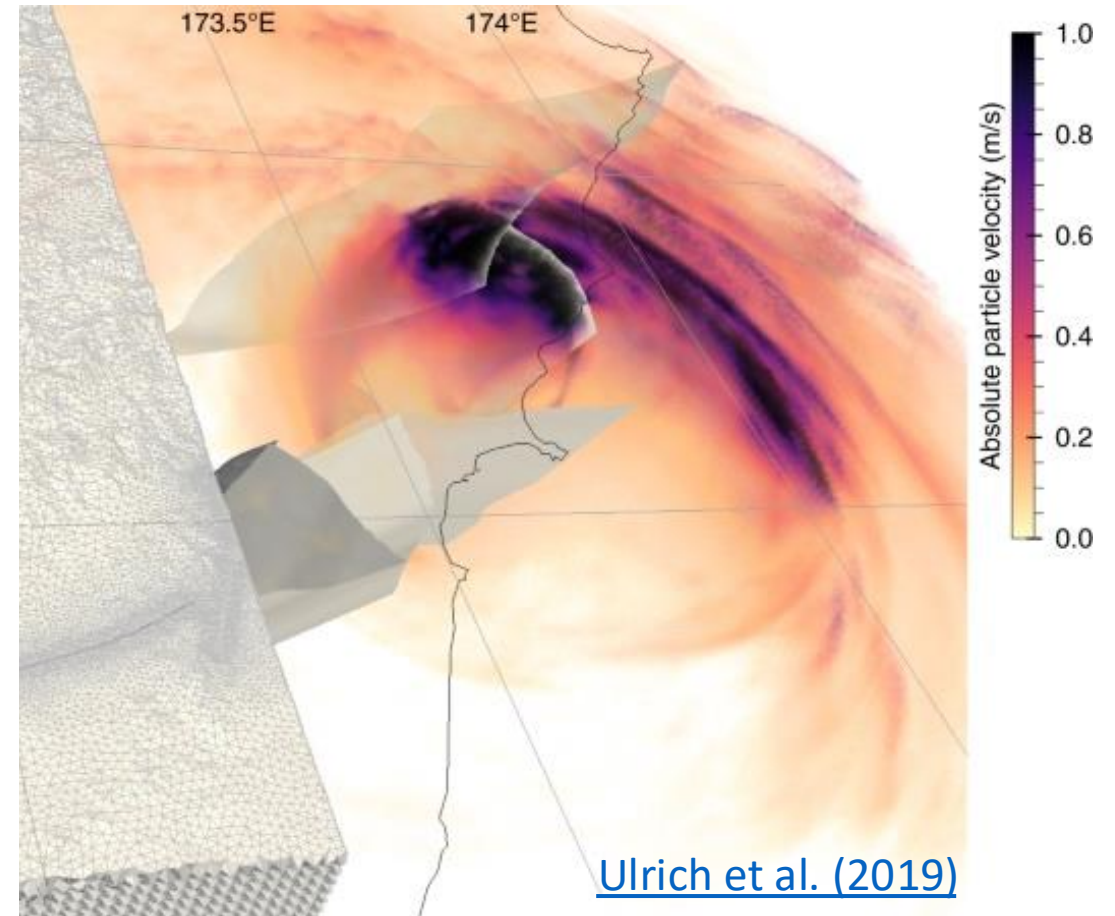
*almost all



What you will learn:

In this course we will take concepts such as mass, force and deformation that are familiar from **point systems** or simple spring and dash-pot models and extend them to **continuum bodies** such as solids and fluids.

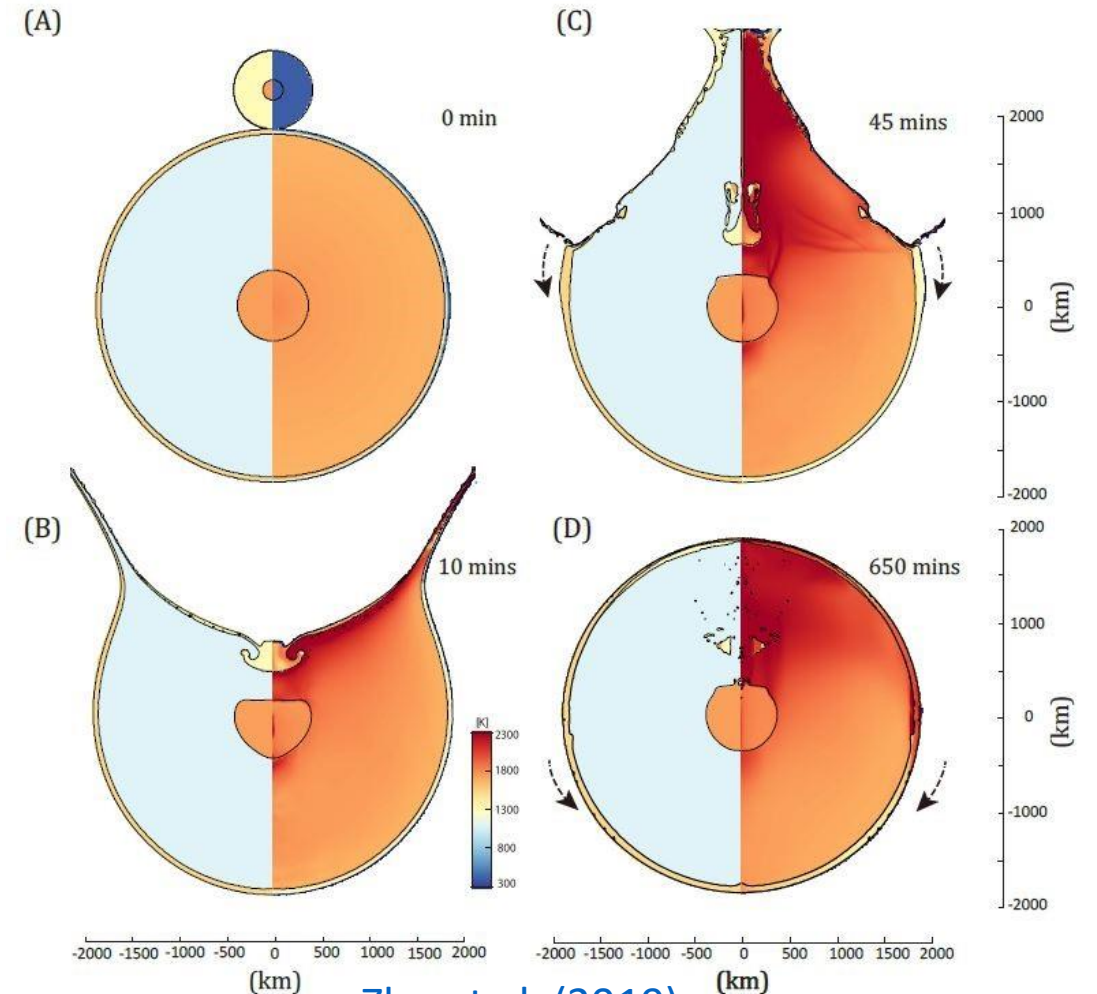
This will allow you to understand the governing and constitutive equations of **modern computational models** and enable you to use and interpret them correctly.



What you will be doing:

Topics:

- Review (3 lec)
- Tensors and stress (9 lec)
- Kinematics and strain (5 lec)
- Constitutive theory (2 lec)
- Conservation laws (2 lec)
- Fluid mechanics (3 lec)
- Solid mechanics (3 lec)



[Zhu et al. \(2019\)](#)

Course website(s)

- Class website: https://mhesse.github.io/continuum_mechanics/
Post all notes, lecture, and recordings (main resource)
- Canvas: <https://utexas.instructure.com/courses/1396276>
(Post and submit homeworks, post grades)
- Piazza: <https://piazza.com/class/mebnfx4v4bmpx>
(class discussion, has latex editing!)

Office hours

Office hours are integral part of the class!

- Conceptual questions answered
- Help with manipulations
- Discuss broader context

⇒ Students that come to office hours do better!

We need to find time for office hours:

Complete poll: <https://www.when2meet.com/?31686602-ktH3O>

Assessment

- Weekly homeworks – due Thursday in class (9:30am)
- Midterm exam on Oct 7 in class.
- Final exam on Dec 9 in class.
- Graduate students project:
- Complete class evaluation (CE)!
- Grade breakdown:

Undergraduates: 50% Homework + 30% Exams + 20% Attendance

Graduate students: 50% Homework + 30% Exams + 20% Project

Graduate class project.

- Aim is to generate interesting up-to-date homework problems that apply class concepts for future classes!
- It is a HW problem so it should not be too complicated!
- Should apply some concept learned in class.
- Short individual meetings weeks 6-8 to discuss potential topics
- Decide on topic in weeks 16-18.
- Projects due before Thanksgiving
- Present/Discuss/Work through Last day of class

Past graduate student projects

Aim:

1. Generate new interesting homework problems that apply class materials to Earth Science problems.

Example 1: Homework problem

PROBLEM SET: Stress on a Fault, Nankai

The Nankai Trough Seismogenic Zone Experiment (NanTroSEIZE), a scientific drilling project conducted by the IODP, has spent over a decade drilling, sampling, imaging, and instrumenting the Nankai subduction zone, offshore Japan, with one of the primary goals of quantifying stress conditions along the plate boundary (Figure 1). The Nankai Trough is created by the subduction of the Philippine Sea plate beneath the Eurasian plate and is located southwest of the Kii Peninsula, Japan. Historically, the Nankai Trough has hosted magnitude 8.0 and greater seismic events and is one of the most tectonically active regions on Earth.

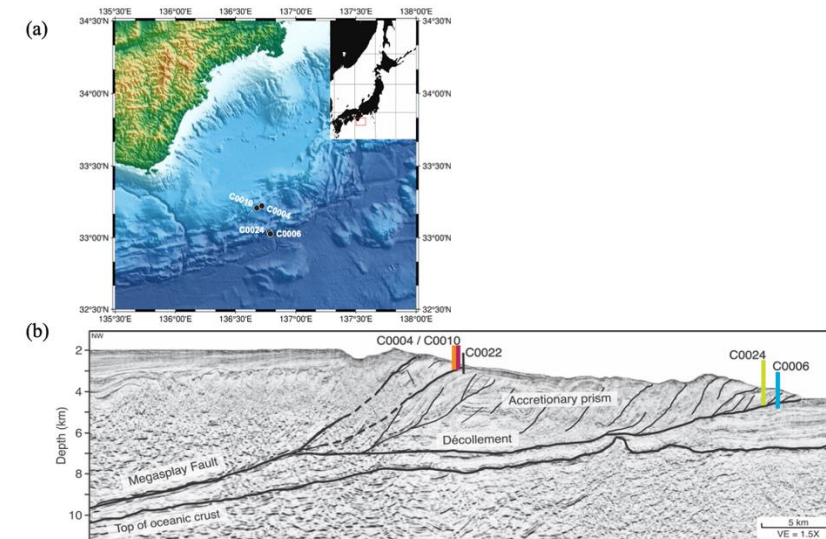


Figure 1: (a) Site map of the IODP NanTroSEIZE drilling sites, southeast Japan. (b) Seismic cross section showing the Nankai subduction zone, accretionary wedge. Site C0004 is highlighted in orange, and the borehole crosses the Megasplay thrust fault.

Past graduate student projects

Aim:

1. Generate new interesting homework problems that apply class materials to Earth Science problems.

Example 2: Homework problem

Homework Question Project for Continuum Mechanics

Hunter Reeves

1. Jakobshavn Glacier on the southwestern flank of Greenland is the world's fastest moving glacier where glacial speed has known to reach up to about 17 km/yr (Joughin et al., 2014).

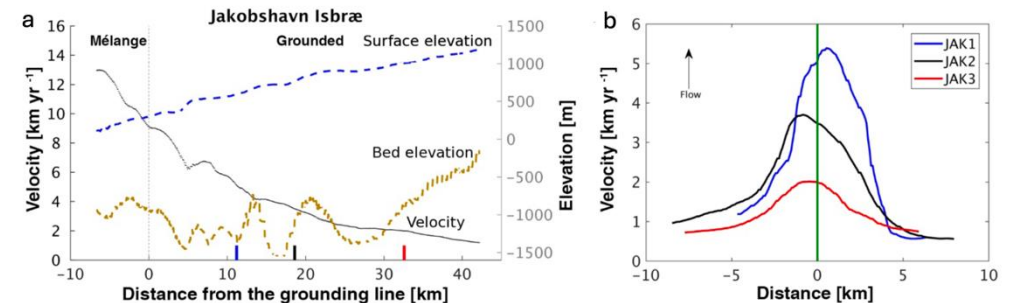


Figure 1. a) This plot shows the velocity, bed elevation, and surface elevation for Jakobshavn glacier from a side view. b) This plot shows the flow of Jakobshavn from a top-down view. Modified from (Lemos et al., 2018).

Glen's Flow Law: $\dot{\epsilon}_{ij} = A\tau_{ij}^n$, where $\dot{\epsilon}_{ij}$ is the strain rate, A is the empirical rate factor, τ_{ij} is the shear stress and n is the flow law exponent.

Past graduate student projects

Aim:

1. Generate new interesting homework problems that apply class materials to Earth Science problems.
2. Opportunity to dive deep and think about the how continuum mechanics applies to your research.

Example 3: Deep dive

2.1. For the fully anisotropic case

"Fully anisotropic" refers to material that has no symmetry constraint on its property or structure. However, we can still eliminate some coefficients by the symmetry of stress and strain tensor themselves:

$$\begin{cases} \sigma_{ij} = \sigma_{ji} \\ \epsilon_{ij} = \epsilon_{ji} \end{cases} \quad (2.1)$$

by substituting Equation (2.1) to (1.3), we have:

$$\begin{cases} C_{ijkl} = C_{jikl} \\ C_{ijkl} = C_{ijlk} \end{cases} \quad (2.2)$$

Then, by expanding the k and l in Equation (1.3) and imposing Equation (2.2):

$$\sigma_{ij} = C_{ij11}\epsilon_{11} + C_{ij22}\epsilon_{22} + C_{ij33}\epsilon_{33} + 2C_{ij12}\epsilon_{12} + 2C_{ij23}\epsilon_{23} + 3C_{ij31}\epsilon_{31} \quad (2.3)$$

This reduced the number of independent coefficients to $6 \times 6 = 36$.

To further reduce the number, we should consider the energy conservation principle, which ensures the elastic energy is independent of the path of loading. The strain energy ψ is given by:

$$\psi = \frac{1}{2} \sigma_{ij} \epsilon_{ij} \quad (2.4)$$

Therefore, we can obtain:

$$\sigma_{ij} = \frac{\partial \psi}{\partial \epsilon_{ij}} = C_{ijkl} \epsilon_{kl} \quad (2.5)$$

$$\frac{\partial \psi}{\partial \epsilon_{ij} \partial \epsilon_{mn}} = \frac{\partial}{\partial \epsilon_{mn}} (C_{ijkl} \epsilon_{kl}) = C_{ijkl} \delta_{km} \delta_{ln} = C_{ijmn} \quad (2.6)$$

Since

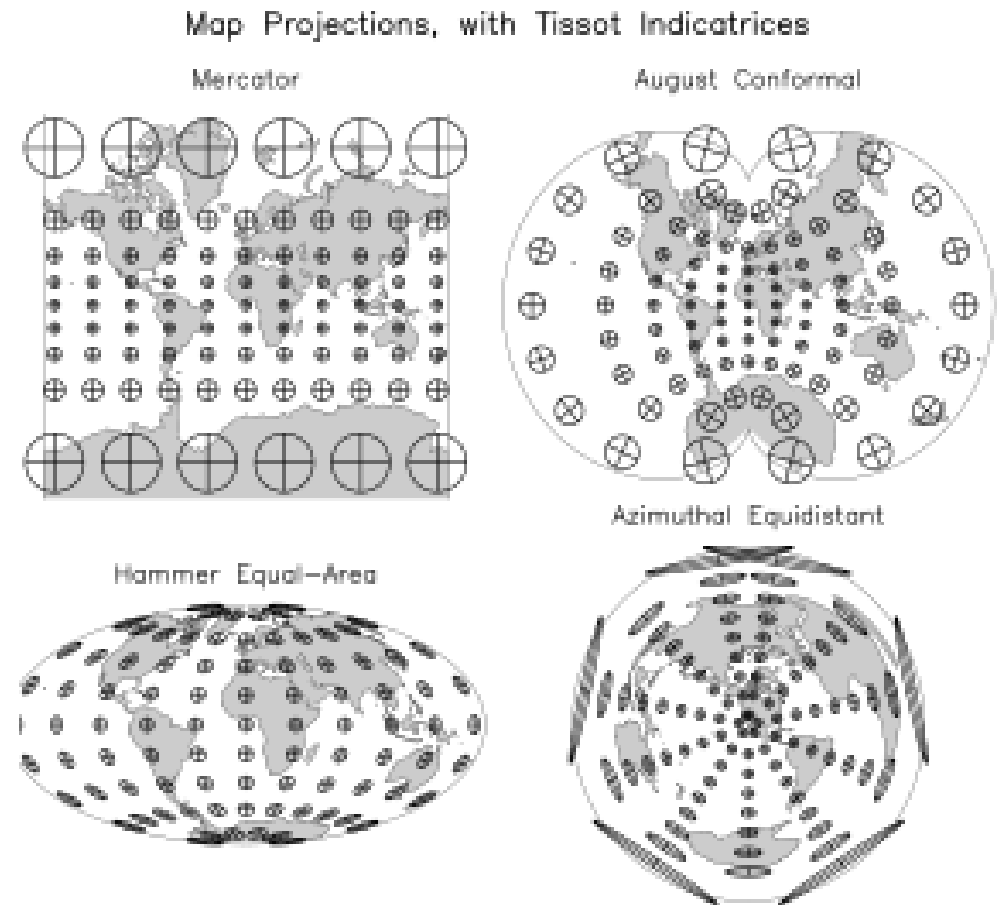
$$\frac{\partial \psi}{\partial \epsilon_{ij} \partial \epsilon_{mn}} = \frac{\partial \psi}{\partial \epsilon_{mn} \partial \epsilon_{ij}} \quad (2.7)$$

and by changing the index mn to kl we have

Future graduate student project

Finite Strain in Cartography

1. Projecting the surface of a sphere onto a plane requires a finite inhomogeneous deformation!
2. The Tissot's Indicatrix is a 'strain ellipse' that illustrates the distortion inherent in all map projections.



Homeworks

- Homework is paper and pencil work.
- Homework needs to be submitted electronically as **PDF files**.
(please don't submit photos of paper sheets)
- If your PDF is very large (MB's), please compress it before submitting
(https://www.ilovepdf.com/compress_pdf)
- You need to **start early** to be successful.
- Late penalty is -10%
- Resubmit **up to 3 PS's** for better grade,
if you have attended **all** relevant classes!

Please don't do this 😊

$$\begin{pmatrix} A_{11,1} & 0 \\ A_{22,2} & 0 \\ 0 & A_{33,3} \end{pmatrix}$$

$\rightarrow H$

$$\nabla_{\underline{x}} \cdot \nabla_{\underline{A}} \underline{x} \rightarrow (\delta_{ij}) e_i$$

$$x_{i,j} A_{i,i} x_{i,i} (e_i \otimes e_j) \cdot (e_i \otimes e_j)$$

$$((e_i \otimes c_j) \cdot c_j) e_i$$

$$\delta_{ij} e_i e_j = e_j e_j = \delta_{jj} = 3$$

Artificial Intelligence



Why?

You are supposed to learn
to think for yourselves!

Exercise your brain.

Collaboration policy

- You can discuss the homework problems.
- You must write up your own solution on your own (cannot copy!)
- Cheating penalty:

I catch you once

-> ask you to work through the HW problem after class

-> if you cannot do the problem -> half points off HW

I catch you again

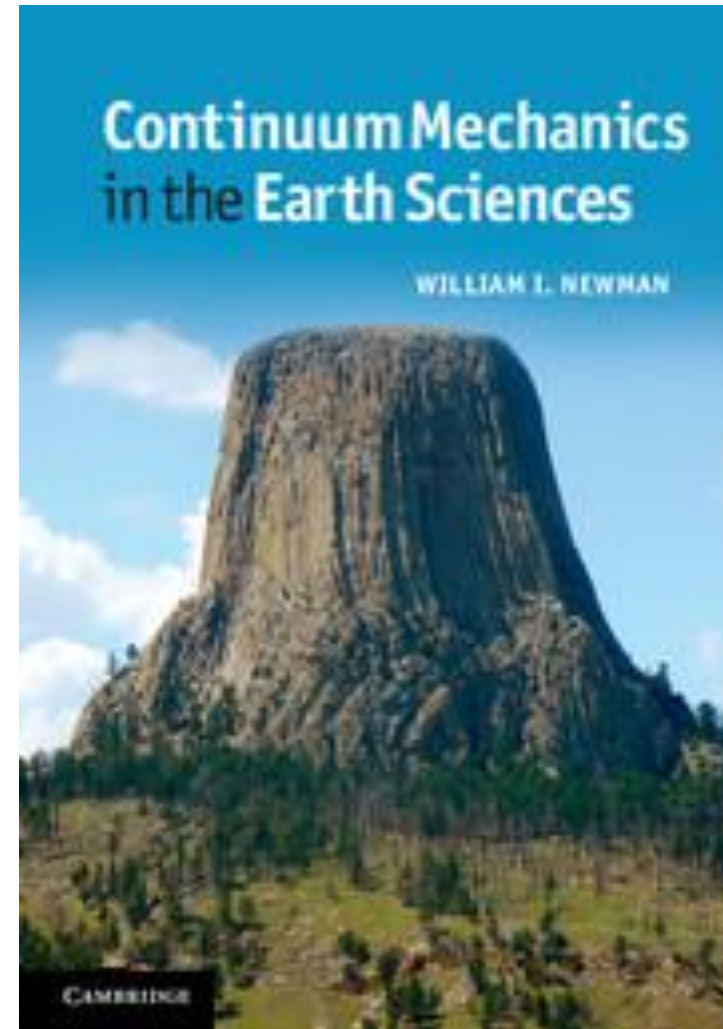
-> 0 points on HW

-> I'll report you to student affairs.

Attendance

- Class attendance is **mandatory** for undergraduates.
- Attending class leads to better results!
- For undergraduates I will take attendance => part of your grade.
- Students that don't attend class are lost on Problem Sets
- If you missed class you **cannot** resubmit PS (also grad students)

Books for class



Read lecture notes not the internet

$$\rho \dot{\underline{v}} = \rho \frac{\partial \underline{v}}{\partial t} + \rho (\nabla_{\underline{x}} \underline{v}) \underline{v} = \frac{\partial}{\partial t} (\rho \underline{v}) - \frac{\partial \rho}{\partial t} \underline{v} + (\nabla_{\underline{x}} \underline{v}) (\rho \underline{v})$$

using mass balance $\frac{\partial \rho}{\partial t} = -\nabla_{\underline{x}} \cdot (\rho \underline{v})$

$$\rho \dot{\underline{v}} = \frac{\partial}{\partial t} (\rho \underline{v}) + \nabla_{\underline{x}} \cdot (\rho \underline{v}) \underline{v} + (\nabla_{\underline{x}} \underline{v}) (\rho \underline{v})$$

using $\nabla \cdot (\underline{a} \otimes \underline{b}) = (\nabla \underline{a}) \underline{b} + \underline{a} \nabla \cdot \underline{b}$ (see HW5 Q4)

$$\rho \dot{\underline{v}} = \frac{\partial}{\partial t} (\rho \underline{v}) + \nabla_{\underline{x}} \cdot (\rho \underline{v} \otimes \underline{v})$$

Hence we have conservative local Eulerian form

$$\frac{\partial}{\partial t} (\rho \underline{v}) + \nabla_{\underline{x}} \cdot (\rho \underline{v} \otimes \underline{v} - \underline{\underline{\underline{\sigma}}}) = \rho \underline{b}$$

conserved quantity: $\rho \underline{v}$ = linear momentum

advective mom. flux: $\rho \underline{v} \otimes \underline{v}$

diffusive mom. flux: $-\underline{\underline{\underline{\sigma}}}$

Lecture notes are main reference.
(G&S is second reference)

Notes often directly help with HW!

You need to use class notation.

